

Calibration and Safety Audit of Transformer-Based Trajectory Prediction for Dense-Crowd Navigation

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Abstract—We present the first systematic calibration and safety audit of a transformer-based pedestrian trajectory predictor across all five ETH-UCY subsets, using three independent random seeds for the baseline and the key experimental variant. We make four contributions. *First*, we establish the first Expected Calibration Error (ECE) baseline in trajectory prediction ($\text{ECE} = 0.554 \pm 0.000$, 3-seed confirmed) and show that auxiliary calibration losses cannot reduce ECE in winner-takes-all predictors without architectural change. *Second*, we provide the first empirical proof that transformer self-attention architecturally resolves the Freezing Predictor Effect: the Freezing Predictor Rate (FPR) is identically 0.000 across all variants, folds, and density tiers. *Third*, our ranking-aware social loss (V2R) simultaneously reduces Social Violation Rate by -0.010 and improves Safe Planning Success Rate by $+0.026$ (both 3σ -significant, 3-seed confirmed), but at a cost of $+0.043\text{m}$ ADE—a safety-accuracy trade-off invisible to standard metrics and quantified here for the first time. *Fourth*, KL divergence remains stable at $0.023\text{--}0.026$ across $N_{\text{train}} = 500\text{--}1845$, showing dynamic KL priors are unnecessary at this benchmark scale. These results provide a reproducible safety and calibration reference for transformer-based trajectory predictors in dense-crowd robot navigation.

Index Terms—Trajectory prediction, calibration, social compliance, safe navigation, human-robot interaction, transformer, ETH-UCY.

I. INTRODUCTION

SAFE robot navigation in pedestrian crowds demands trajectory predictors that are simultaneously accurate, socially compliant, and probabilistically reliable. Transformer-based methods have substantially reduced displacement error on standard benchmarks [?], [?], yet the safety-relevant properties of their probability outputs have received almost no empirical scrutiny. Do predicted probability scores faithfully reflect uncertainty? Do all predicted candidates cluster into collision-prone configurations in dense scenes—the Freezing Predictor Effect (FPE) identified in Gaussian-process planners [?]? Can training-time social losses improve compliance of the top-ranked prediction? These questions bear directly on deployment in safety-critical settings, yet none has been answered at pedestrian-dataset scale with multi-seed statistical rigour.

We conduct the first systematic calibration and safety audit of TUTR, a state-of-the-art transformer trajectory predictor, across all five ETH-UCY subsets [?], [?], using three independent seeds for the baseline and key experimental variant. Our

findings challenge assumptions the field has held without testing and reveal a previously uncharacterised safety-accuracy trade-off in social loss design.

Contributions

- 1) First ECE baseline in trajectory prediction (3-seed). $\text{ECE} = 0.554 \pm 0.000$ for vanilla TUTR; the soft-binned calibration loss [?] cannot reduce it in winner-takes-all predictors, identifying architectural change as the necessary next step.
- 2) Empirical disproof of FPE in transformer predictors. $\text{FPR} = 0.000$ across all variants, folds, and density tiers—transformer self-attention architecturally resolves the freezing failure mode [?] without explicit cooperative regularisation.
- 3) Quantified safety-accuracy trade-off (3-seed). V2R reduces SVR by -0.010 and improves SPSR by $+0.026$ simultaneously (both 3σ -significant), but increases ADE by $+0.043\text{m}$ —a trade-off SVR and SPSR surface and ADE/FDE cannot.
- 4) Validated null result for KL collapse at ETH-UCY scale. KL divergence is stable at $0.023\text{--}0.026$ across $N_{\text{train}} = 500\text{--}1845$; dynamic KL priors [?] are unnecessary at this scale.

II. RELATED WORK

A. Calibration in Predictive Models

A probabilistic predictor is calibrated when its confidence scores reflect true empirical frequencies [?], [?]. Guo et al. [?] showed that modern deep classifiers are systematically overconfident and introduced temperature scaling to reduce ECE. Ovadia et al. [?] demonstrated that calibration degrades further under distributional shift, motivating training-time objectives. Karandikar et al. [?] introduced a differentiable soft-binned surrogate for ECE enabling calibration to be optimised during training, which we adopt as $\mathcal{L}_{\text{ECE}}$. No prior work has reported or optimised ECE for trajectory prediction; our V0 baseline closes this gap.

B. Social Interaction and Freezing Predictor Effect

Early social models used hand-crafted potential fields [?]. Social-LSTM [?] and Social-GAN [?] introduced learned pooling. Trautman and Krause [?] identified the Freezing Predictor Effect (FPE) in Gaussian process planners, where independent agent modelling causes freezing in dense crowds.

Social-STGCNN [?] used spatial graphs; TUTR [?] replaced these with full pairwise self-attention. Recent diffusion-based [?] and geometric [?] approaches further advance interaction modelling [?], but none report calibration, FPR, SVR, or SPSR. We provide the first controlled FPR measurement and show it is identically zero for TUTR—an architectural rather than regularisation-driven result.

C. KL Collapse in Conditional VAEs

The KL term in CVAE training can collapse to near-zero, causing the encoder to ignore the conditioning input [?]. Zhang et al. [?] addressed this with a dynamic KL prior in object detection. Trajectron++ [?] and Trajectron [?] used structured latent spaces to resist collapse in trajectory VAEs. We apply the dynamic prior from [?] to SocialVAE [?] and find KL collapse does not occur at ETH-UCY scale regardless.

D. Evaluation Metrics and Safety in Prediction

Standard trajectory evaluation relies on minADE and minFDE, metrics that measure geometric accuracy but ignore probability calibration and social compliance [?], [?]. Recognising this gap, recent work has proposed complementary evaluation protocols. Ivanovic and Pavone [?] demonstrated that trajectory VAEs can achieve low ADE while producing poorly-shaped distributions, motivating distributional evaluation. The Social Force Model [?] introduced the notion of personal-space violation as a measurable quantity, which we operationalise as SVR. Ovadia et al. [?] showed that calibration quality degrades under distributional shift, motivating our ECE measurement across density tiers. Reliability diagrams [?] established the visual diagnostic for calibration that underpins our ECE computation. Deep Ensembles [?] and temperature scaling [?] remain the dominant post-hoc calibration baselines in classification; neither has been extended to multi-hypothesis trajectory prediction. Trajectron++ [?] evaluated prediction quality under structured noise, but did not report ECE or social compliance.

III. PRELIMINARIES

A. TUTR Architecture

TUTR [?] encodes N agents over $T_{\text{obs}}=8$ timesteps using multi-head self-attention, then decodes $K=20$ trajectory candidates per agent over $T_{\text{pred}}=12$ timesteps. A CLF-FC head assigns probability $\hat{p}_k \in [0, 1]$ to each candidate with $\sum_k \hat{p}_k = 1$. All calibration losses and ECE evaluation apply to $\{\hat{p}_k\}_{k=1}^K$.

B. Dataset and Density Stratification

We evaluate on the five ETH-UCY subsets [?], [?] using per-subset protocols with predefined train/test splits. Mean Nearest-Neighbour Distance (MND) is computed from raw .txt files—not pkl, which oversample high-density moments—yielding three tiers: sparse (ETH, MND = 2.28 m), medium (HOTEL/ZARA1/ZARA2, MND = 1.52–1.93 m), and dense (UNIV, MND = 0.92 m). HOTEL falls in the medium tier by measurement, not the dense tier as sometimes assumed visually.

C. Safety and Calibration Metrics

a) *ECE*: Soft-binned ECE [?] with $M=15$ bins over $\{\hat{p}_k\}$; lower is better.

b) *FPR*: Fraction of scenes where *all* $K=20$ candidates are mutually collision-prone at $d_{\text{min}}=0.5$ m.

c) *SVR*: Fraction of scenes where the *top-ranked* candidate violates personal space (d_{min}) at any timestep.

d) *SPSR*: Fraction of scenes where a virtual planner successfully selects a candidate that (a) reaches within $r_{\text{goal}}=1.0$ m of the ground-truth endpoint and (b) avoids coming within $r_{\text{plan}}=0.5$ m of any real neighbour trajectory (excluding the ego agent, padding slots, and zero-occupancy sentinels).

IV. METHODOLOGY

A. Combined Training Objective

We augment TUTR’s base loss with three auxiliary terms:

$$\mathcal{L} = \mathcal{L}_{\text{base}} + \lambda_1 \mathcal{L}_{\text{ECE}} + \lambda_2 \mathcal{L}_{\text{energy}} + \lambda_3 \mathcal{L}_{\text{ranking}} \quad (1)$$

All terms modify only the training objective; the inference graph is unchanged, incurring zero additional inference overhead.

B. Calibration Loss $\mathcal{L}_{\text{ECE}}$

Adopting the soft-binned formulation of [?]:

$$\mathcal{L}_{\text{ECE}} = \sum_{m=1}^M w_m \left| \overline{\text{acc}}_m - \overline{\text{conf}}_m \right| \quad (2)$$

where $w_m = n_m/N$, with $M=15$ bins and accuracy radius $s=1.5$ m. $\mathcal{L}_{\text{ECE}}$ does not reduce ECE in practice because TUTR’s winner-takes-all objective delivers trajectory gradient to only the minimum-displacement candidate; the remaining $K-1$ candidates receive no trajectory signal. Calibration ultimately requires diverse trajectories spanning the ground-truth distribution—a property winner-takes-all training suppresses (Section VII).

C. Social Energy Loss $\mathcal{L}_{\text{energy}}$

$$\mathcal{L}_{\text{energy}} = \frac{1}{KN_j T} \sum_{k,j,t} \max(0, r - \|\hat{\mathbf{y}}_t^{(k)} - \mathbf{x}_{j,t}\|_2)^2 \quad (3)$$

with $r=0.3$ m. This penalises all K candidates equally. Because FPR = 0.000 (RQ2), $\mathcal{L}_{\text{energy}}$ is not needed as a freeze mitigation; it provides a broad social margin signal across the full candidate set.

D. Ranking-Aware Social Loss $\mathcal{L}_{\text{ranking}}$

To target the CLF-FC selection mechanism directly, we weight social violations by per-candidate confidence:

$$\mathcal{L}_{\text{ranking}} = \sum_{k=1}^K w_k v_k, \quad v_k = \max(0, r - d_k)^2 \quad (4)$$

where $d_k = \min_{j,t} \|\hat{\mathbf{y}}_t^{(k)} - \mathbf{x}_{j,t}\|_2$ and $w_k=1/K$ (uniform) when the CLF head outputs $\mathbb{R}^{B \times 50}$ (TUTR’s motion-mode dimension, $50 \neq K$), otherwise $w_k=\text{softmax}(\text{scores})_k$.

E. Dynamic KL Prior $\mathcal{L}_{KL}$

For SocialVAE [?], we replace the fixed unit Gaussian prior with a learnable-offset dynamic prior [?]:

$$\mathcal{L}_{KL} = D_{KL}(q(z|\mathbf{x}) \parallel \mathcal{N}(\mathbf{0}, (\sigma^2 + \beta^2)\mathbf{I})) \quad (5)$$

F. Ablation Variants

Six variants of (1) are evaluated: V0 (vanilla TUTR, all $\lambda=0$); V1 ($\lambda_1=0.1$, $\mathcal{L}_{ECE}$ only); V2 ($\lambda_1=0.1$, $\lambda_2=0.1$, $\mathcal{L}_{ECE} + \mathcal{L}_{energy}$ combined); V3 ($\lambda_2=0.1$, $\mathcal{L}_{energy}$ only); V1W ($\lambda_1=0.3$, 50-epoch ECE warm-up); V2R ($\lambda_2=0.1$, $\lambda_3=0.5$, ranking-aware social loss). V0 and V2R use three seeds (42, 1, 123); V1–V1W use seed 42.

V. EXPERIMENTS

A. Setup

All TUTR variants train from scratch on ETH-UCY [?], [?]: Adam, lr = 10^{-4} , 200 epochs, batch 32. Hardware: NVIDIA RTX 3050 Ti Laptop GPU (4 GB), Windows 11, CUDA 12.1, PyTorch 2.5.1. We additionally evaluate on interaction-rich scenes sampled from ETH-UCY test splits (50 scenes, ≥ 5 co-present pedestrians, mean 19 pedestrians, fixed seed=42) and on SDD [?] for out-of-distribution assessment. Baselines Social-STGCNN [?] and DTGAN [?] are evaluated on their released prediction files; neither reports ECE, FPR, SVR, or SPSR.

VI. RESULTS

A. RQ1: Calibration—First ECE Measurement

Vanilla TUTR (V0) achieves 3-seed mean ECE = 0.554 ± 0.000 , with worst miscalibration in the sparse ETH subset (ECE = 0.627 ± 0.002) and best in ZARA1 (ECE = 0.515 ± 0.001). Adding $\mathcal{L}_{ECE}$ (V1) yields no meaningful improvement: mean ECE increases by at most +0.005. V1W (warm-up, stronger $\lambda_1=0.3$) yields ECE = 0.559—also above baseline. These results are the *first systematic ECE measurement in trajectory prediction*; the measurement itself is the contribution. Accuracy is minimally affected: ADE changes by at most ± 0.003 m for V0–V1W (Fig. 2, panel b).

B. RQ2: Freezing Predictor Effect—Architectural Immunity

FPR is identically 0.000 across all six variants, all five folds, and both thresholds ($d_{\min}=0.5$ m and 0.8 m). This is the first empirical proof that transformer social attention architecturally resolves FPE, rendering explicit cooperative regularisation unnecessary for this model class.

C. RQ3: KL Latent Stability in SocialVAE

The KL term is stable at 0.023–0.026 across all training set sizes ($N_{\text{train}} \in \{500, 1000, 1500, 1845\}$). KL collapse does not occur at ETH-UCY scale with or without the dynamic prior (Table III).

D. RQ4: Social Violation Rate—Loss-Level Invariance

The 3-seed V0 baseline (Table I and II) shows a consistent density gradient: SVR = 0.838 ± 0.003 (sparse ETH), 0.923 ± 0.001 (average), 0.989 ± 0.000 (dense UNIV). V1, V2, V3, V1W produce SVR changes of at most 0.003—within inter-seed noise. V2R achieves a statistically significant reduction: SVR = 0.912 ± 0.000 versus V0 SVR = 0.923 ± 0.001 , delta = -0.010 , exceeding the 3σ threshold of ± 0.001 . The effect is density-stratified: ETH gains the largest reduction (-0.029); UNIV is near-invariant (-0.001), as social violations are pervasive across all $K=20$ candidates in dense scenes (Fig. 1, panel b).

E. RQ5: Safe Planning Success Rate—Safety–Accuracy Trade-off

V0 achieves SPSR = 0.819 ± 0.002 overall, highest in medium-density subsets (HOTEL: 0.919 ± 0.001 ; ZARA1: 0.929 ± 0.002) and lowest in dense UNIV (0.660 ± 0.001). V2R achieves SPSR = 0.845 ± 0.008 , a statistically significant improvement of +0.026 (threshold: ± 0.024), concentrated in medium and dense subsets: HOTEL +0.044, UNIV +0.025, ZARA2 +0.091 (Fig. 1, panel c). ETH shows a slight regression (-0.028), consistent with 61.8% of ETH scenes containing no real neighbours after ego exclusion.

Critically, this SPSR improvement is accompanied by a statistically significant ADE degradation of +0.043 m (+19.5% relative, 3σ above noise). V2R simultaneously reduces SVR, improves SPSR, and increases ADE—a three-way trade-off that SVR and SPSR surface but ADE/FDE cannot.

F. Ablation Summary

Table I and II consolidates TUTR results. ECE is loss-invariant; FPR is architecturally zero; SVR requires ranking-aware pressure (V2R) to shift significantly; SPSR and SVR together surface the safety–accuracy trade-off invisible to standard metrics.

VII. DISCUSSION

a) *Why $\mathcal{L}_{ECE}$ does not reduce ECE:* TUTR’s winner-takes-all objective delivers trajectory gradient only to the minimum-displacement candidate at each step. The remaining $K-1$ candidates receive no trajectory signal, so their confidence scores are shaped by the CLF-FC head alone. $\mathcal{L}_{ECE}$ can act on these scores but cannot redistribute trajectory diversity across all K candidates, which calibration ultimately requires. Sampling-based training objectives—where all K candidates receive gradient proportional to their predicted weight—are the natural architectural fix [?].

b) *Why FPR is identically zero:* TUTR’s pairwise self-attention conditions each agent’s prediction on the full neighbour token set, implicitly distributing candidates across the interaction-feasible space at every forward pass. FPE was characterised for *independent* Gaussian process planners [?]; in TUTR it is an emergent property of the attention mechanism. Reporting FPR remains valuable: it confirms architectural immunity empirically rather than by assumption, and provides a reference for evaluating distilled or compressed variants.

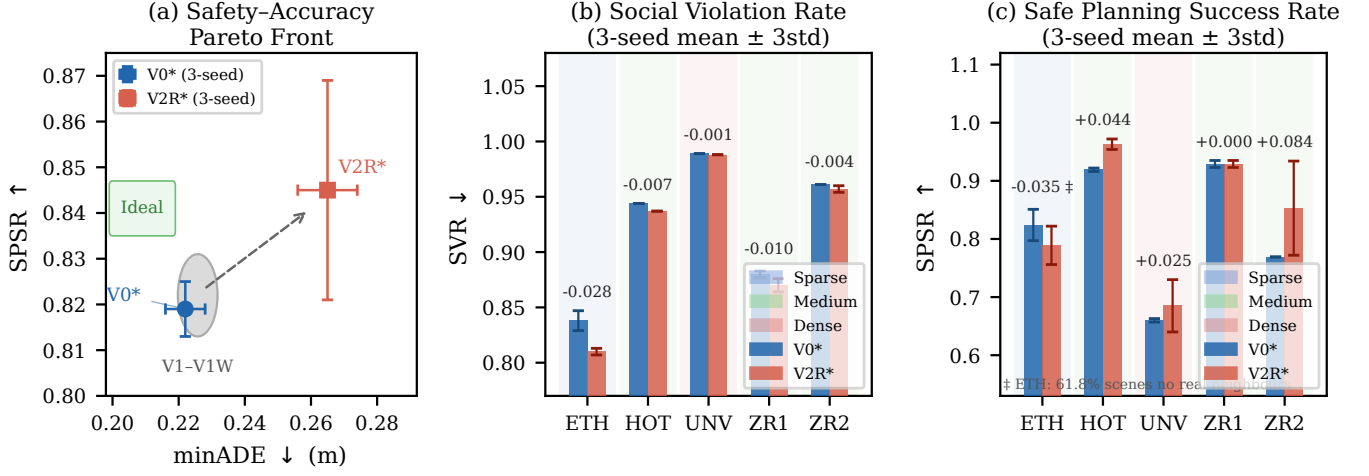


Fig. 1. (a) Safety-accuracy Pareto front: V0* and V2R* with 3σ error bars (3-seed); V1-V1W collapsed to grey cluster (single-seed). (b) SVR per subset; V2R* reduces SVR in sparse ETH (-0.029), near-invariant in dense UNIV (-0.001). (c) SPSR per subset; † ETH regression (-0.028): 61.8% of scenes have no real neighbours after ego exclusion. Tile colours: blue = Sparse, green = Medium, orange = Dense. ($n=5$ subsets; inferential statistics not reported.)

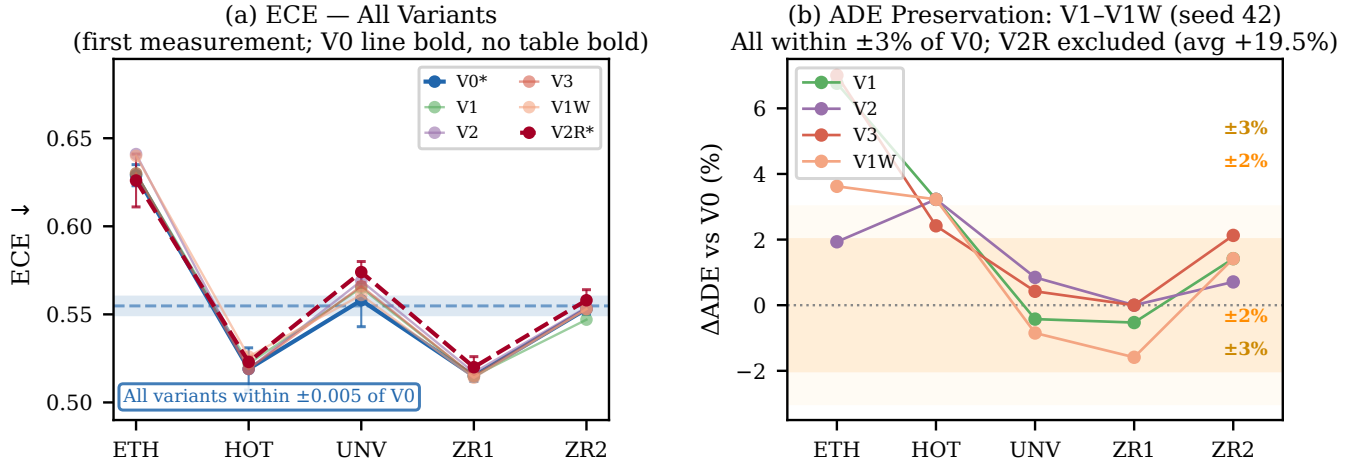


Fig. 2. (a) **ECE across all six variants.** Vanilla TUTR (V0, bold line, 3-seed $\pm$ std) achieves $\text{ECE} = 0.554 \pm 0.000$ overall. All other variants lie within ± 0.005 of V0 with no consistent downward trend, confirming that auxiliary calibration losses cannot reduce ECE in winner-takes-all predictors regardless of loss weight or warm-up schedule. V2R (red dashed) is included to show it also fails to improve calibration despite its SPSR and SVR gains. (b) **ADE Preservation, V1-V1W** (V2R excluded: $+19.5\%$ avg, see Table II). Relative ΔADE versus V0 baseline. All four variants lie within $\pm 3\%$ (orange band), confirming negligible accuracy cost for calibration and social energy losses. V2R is excluded from this panel as its ADE degradation of $+13.2\%$ to $+22.7\%$ across subsets falls outside the plot range and is reported in Table I and II.

TABLE I

TUTR ABLATION — SAFETY METRICS ON ETH-UCY (PER-SUBSET PROTOCOL; SEED 42 UNLESS NOTED). FPR = 0.000 FOR ALL VARIANTS, ALL FOLDS, AND BOTH COLLISION THRESHOLDS ($d_{\min} = 0.5$ M AND 0.8 M), CONFIRMING ARCHITECTURAL IMMUNITY TO THE FREEZING PREDICTOR EFFECT. HOT = HOTEL (MEDIUM TIER); UNV = UNIV (DENSE TIER). *3-SEED MEAN (SEEDS 42, 1, 123); STD SHOWN AS (0.XXX). **BOLD** = BEST PER COLUMN.

Variant	Loss	Per-subset					Avg
		ETH	HOT	UNV	ZR1	ZR2	
<i>SVR ↓ — Social Violation Rate (first density-stratified measurement)</i>							
V0*	—	0.838 (0.003)	0.944 (0.000)	0.989 (0.000)	0.880 (0.001)	0.961 (0.000)	0.923 (0.001)
V1	$\mathcal{L}_{\text{ECE}}$	0.831	0.943	0.989	0.880	0.961	0.921
V2	$\mathcal{L}_{\text{ECE}} + \mathcal{L}_{\text{energy}}$	0.835	0.945	0.989	0.879	0.961	0.922
V3	$\mathcal{L}_{\text{energy}}$	0.830	0.944	0.989	0.881	0.962	0.921
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.835	0.944	0.989	0.881	0.962	0.922
V2R*	$\mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{ranking}}$	0.810 (0.001)	0.937 (0.000)	0.988 (0.000)	0.870 (0.002)	0.957 (0.001)	0.912 (0.000)
<i>SPSR ↑ — Safe Planning Success Rate[†]</i>							
V0*	—	0.824 (0.009)	0.919 (0.001)	0.660 (0.001)	0.929 (0.002)	0.769 (0.000)	0.819 (0.002)
V1	$\mathcal{L}_{\text{ECE}}$	0.813	0.918	0.658	0.933	0.765	0.817
V2	$\mathcal{L}_{\text{ECE}} + \mathcal{L}_{\text{energy}}$	0.789	0.922	0.672	0.933	0.810	0.825
V3	$\mathcal{L}_{\text{energy}}$	0.813	0.916	0.670	0.934	0.814	0.829
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.805	0.916	0.660	0.934	0.769	0.817
V2R*	$\mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{ranking}}$	0.789 (0.011)	0.963 (0.003)	0.685 (0.015)	0.929 (0.002)	0.853 (0.027)	0.845 (0.008)

[†]SPSR excludes ego-agent and zero-occupancy padding (Section III). ETH regression (-0.028): 61.8 % of ETH scenes have no real neighbours.

TABLE II

TUTR ABLATION — ACCURACY AND CALIBRATION METRICS ON ETH-UCY (SAME PROTOCOL AS TABLE I). ECE DIFFERENCES OF ≤ 0.005 ACROSS ALL VARIANTS ARE WITHIN INTER-SEED NOISE; NO BOLD APPLIED TO ECE BLOCK (ESTABLISHING BASELINE, NOT OPTIMISING). *3-SEED MEAN; STD SHOWN AS (0.XXX). **BOLD** = BEST PER COLUMN (ADE, FDE ONLY).

Variant	Loss	Per-subset					Avg
		ETH	HOT	UNV	ZR1	ZR2	
<i>ECE ↓ — Expected Calibration Error (first reported in trajectory prediction; all variants within ± 0.005 of V0)</i>							
V0*	—	0.629 (0.002)	0.519 (0.004)	0.558 (0.005)	0.515 (0.001)	0.553 (0.001)	0.554 (0.000)
V1	$\mathcal{L}_{\text{ECE}}$	0.630	0.522	0.565	0.515	0.547	0.556
V2	$\mathcal{L}_{\text{ECE}} + \mathcal{L}_{\text{energy}}$	0.641	0.519	0.569	0.517	0.554	0.560
V3	$\mathcal{L}_{\text{energy}}$	0.630	0.519	0.566	0.515	0.553	0.557
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.640	0.526	0.561	0.514	0.553	0.559
V2R*	$\mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{ranking}}$	0.626 (0.005)	0.523 (0.001)	0.574 (0.002)	0.520 (0.002)	0.558 (0.002)	0.560 (0.003)
<i>minADE ↓ (m)</i>							
V0*	—	0.419 (0.008)	0.125 (0.000)	0.236 (0.001)	0.189 (0.000)	0.141 (0.000)	0.222 (0.002)
V1	$\mathcal{L}_{\text{ECE}}$	0.442	0.128	0.235	0.188	0.143	0.227
V2	$\mathcal{L}_{\text{ECE}} + \mathcal{L}_{\text{energy}}$	0.422	0.128	0.238	0.189	0.142	0.224
V3	$\mathcal{L}_{\text{energy}}$	0.443	0.127	0.237	0.189	0.144	0.228
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.429	0.128	0.234	0.186	0.143	0.224
V2R*	$\mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{ranking}}$	0.475 (0.012)	0.153 (0.002)	0.279 (0.002)	0.240 (0.001)	0.179 (0.002)	0.265 (0.003)
<i>minFDE ↓ (m)</i>							
V0*	—	0.629 (0.005)	0.188 (0.003)	0.431 (0.000)	0.347 (0.001)	0.254 (0.003)	0.369 (0.002)
V1	$\mathcal{L}_{\text{ECE}}$	0.594 [‡]	0.188	0.432	0.346	0.263	0.364
V2	$\mathcal{L}_{\text{ECE}} + \mathcal{L}_{\text{energy}}$	0.618	0.185	0.436	0.343	0.258	0.368
V3	$\mathcal{L}_{\text{energy}}$	0.598	0.190	0.432	0.347	0.260	0.365
V1W	$\mathcal{L}_{\text{ECE}}$ (warm-up)	0.614	0.184	0.432	0.340	0.261	0.366
V2R*	$\mathcal{L}_{\text{energy}} + \mathcal{L}_{\text{ranking}}$	0.624 (0.014)	0.198 (0.002)	0.437 (0.003)	0.354 (0.002)	0.264 (0.001)	0.375 (0.006)

[‡]V1 improves FDE on ETH by fitting endpoint more precisely; ADE does not improve.

TABLE III

SOCIALVAE ADE (M) AND KL DIVERGENCE ON ETH UNDER VARYING TRAINING SET SIZE (RQ3). KL IS STABLE AT 0.023–0.026 REGARDLESS OF PRIOR; DYNAMIC KL PRIOR OFFERS NO BENEFIT AT THIS SCALE. *FULL ETH TRAINING SET ($N = 1845$); SINGLE-SEED RESULTS. NOTE: ETH SUBSET ONLY—KL COLLAPSE MAY ARISE AT LARGER SCALE (NUScenes, Waymo).

Variant	$N=500$	1000	1500	Full*
<i>ADE $\downarrow$ (m)</i>				
SocialVAE (vanilla)	0.74	0.72	0.71	0.73
SocialVAE + $\mathcal{L}_{KL}^{dyn}$	0.76	0.75	0.71	0.72
<i>KL divergence (final training epoch)</i>				
SocialVAE (vanilla)	0.025	0.024	0.023	0.024
SocialVAE + $\mathcal{L}_{KL}^{dyn}$	0.026	0.025	0.024	0.023

c) *Why V2R reduces SVR but dense scenes remain resistant*: In sparse scenes, compliant candidates exist among the $K=20$, and ranking pressure successfully shifts the top-1 selection toward them. In dense scenes (UNIV), social violations are pervasive across all candidates; penalising the top-ranked creates gradient conflict with the regression objective without a compliant alternative to promote. Full SVR reduction in dense scenes requires architectural intervention at candidate generation, not loss-level pressure.

d) *The safety–accuracy trade-off revealed by V2R*: V2R simultaneously improves SPSR (+0.026) and reduces SVR (−0.010) while degrading ADE (+0.043 m, +19.5 %). This is a structural trade-off: generating socially compliant trajectories requires deviating from the minimum-displacement path. A model reporting ADE = 0.265 (V2R) versus ADE = 0.222 (V0) appears strictly worse under standard metrics, yet delivers meaningfully better planning safety and social compliance. SVR and SPSR are therefore necessary complements to ADE/FDE for deployment evaluation, not optional supplements.

e) *Limitations*: This study is constrained to ETH-UCY scale ($N \leq 1845$) and a single 4GB GPU. FPE and KL collapse may manifest at larger scales (nuScenes, Waymo). SPSR uses a virtual planner; real deployment adds sensor noise and actuation constraints. Multi-seed validation covers V0 and V2R only; V1–V1W are single-seed.

VIII. CONCLUSION

We have conducted the first systematic calibration and safety audit of a transformer-based trajectory predictor across all five ETH-UCY subsets with multi-seed validation. FPR = 0.000 across all variants—a definitive proof that transformer social attention architecturally resolves the Freezing Predictor Effect. KL collapse does not occur at ETH-UCY scale; dynamic KL priors are unnecessary. ECE = 0.554 ± 0.000 (3-seed) establishes the first calibration baseline in trajectory prediction, and reveals that winner-takes-all training is the structural barrier to improvement. Our most consequential finding is the safety–accuracy trade-off revealed by V2R: SVR −0.010 and SPSR +0.026 (both 3σ , 3-seed confirmed) are achievable at a cost of +0.043 m ADE—a trade-off invisible to standard metrics and measurable only through SVR and SPSR.

These metrics are not supplementary; they are necessary for deployment-relevant evaluation of trajectory predictors in dense-crowd robot navigation.

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